

SIAI 2026/2027

Exercises #1

Goals

Service configuration

Users and groups

TCP/IP networking

VM inventory

- **DebianGW:** intended to be used as network gateway. Has 2 network interfaces: one on the department network (*ens33*: IP assigned via DHCP) and another one on a private network (*ens34*).
- **DebianSRV:** intended to be used as a client machine. Has 1 network interface, placed on a private network.

Materials

- Gestão de Sistemas e Redes em Linux, Jorge Granjal, FCA, “Capítulo 3. Níveis de execução e Serviços”.
- Gestão de Sistemas e Redes em Linux, Jorge Granjal, FCA, “Capítulo 4. Utilizadores e quotas”.
- Gestão de Sistemas e Redes em Linux, Jorge Granjal, FCA, “Capítulo 5. Configuração TCP/IP”

Other materials

- Group sheet:
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1. Service configuration (GW machine)

- Login with then VM and check which address was assigned on the public interface (via DHCP) and connect to the GW through SSH.
- Change the hostname to *siai-supergw*
- Install the *apache* and *dovecot* services
- List all services available in the system
- Make sure that the previously installed services are started automatically, when the operating system boots up

2. Users and groups (GW machine)

- Create the group *siai* with GID 5000
- Create a personal user in Linux with the same username as your account at student.dei.uc.pt. This user should belong to the group *siai* created previously
- Define the complete name and a password for your new user on the system. Try logging in using the new account.

3. TCP/IP networking (SRV and GW)

SRV: Login and configure the IP address of the network interface which is connected to the internal network to an IP address in the range 192.168.0.0/24. The default gateway should be X.X.0.254.

GW: Configure the IP address of the network interface (*ens34*) which is connected to the internal network to X.X.0.254. Confirm that you can reach the SRV host (via ping).

4. Practical network infrastructure (SRV and GW)

Implement the following network infrastructure. Configure NAT in the GW to ensure connection between SRV and the Internet.

